

epidesc: An R package for computing epidemiological descriptors to characterise infectious disease transmission patterns

Raquel Martins Lana ^{1,2}, Carles Milà Garcia ¹, Diego Ricardo Xavier ², Iasmim Ferreira de Almeida ^{2,3}, Cláudia Torres Codeço ⁴, Daniela Sophie Lührsen ¹, Christovam Barcellos ², and Rachel Lowe ^{1,5}

¹ Barcelona Supercomputing Center, Spain  ² Climate and Health Observatory, ICICT/Fiocruz, Brazil  ³ Getúlio Vargas Foundation, Brazil  ⁴ Scientific Computing Program, Fiocruz, Brazil  ⁵ Catalan Institution for Research & Advanced Studies (ICREA), Spain  Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

The R package `epidesc` provides a standardised pipeline for computing epidemiological descriptors (epi-descriptors) from weekly case count time series of infectious diseases. These epi-descriptors characterise the transmission dynamics, by quantifying the magnitude, duration, and temporal patterns of cases at a spatial unit, enabling the classification of areas into distinct transmission patterns (e.g., persistent, epidemic, episodic, outbreaks, trends). The workflow requires three steps: formatting temporal data into epidemiological year-week identifiers, selecting descriptors and their parameters from a built-in catalogue, and computing descriptors for each spatial unit and epidemiological year. `epidesc` is designed for researchers and public health professionals working on infectious disease surveillance, early warning systems, and spatial epidemiology.

Statement of need

Climate-sensitive diseases such as dengue, malaria, and leptospirosis exhibit heterogeneous transmission dynamics across space and time, driven by climatic, demographic, and socioeconomic factors (Alcayna et al., 2025; Almeida et al., 2022). Studies have demonstrated that the spatio-temporal diffusion of dengue is conditioned by population mobility, vector distribution, and local social and environmental characteristics, producing distinct epidemic patterns even at the neighbourhood scale (Xavier et al., 2017). Understanding and classifying these dynamics is essential for designing targeted surveillance strategies and allocating resources efficiently. Inspired by the epi-features calculated for influenza curves conceived by (Tabataba et al., 2017), Almeida et al. (2022) proposed a set of 13 epi-descriptors to characterise dengue transmission patterns in Brazilian municipalities, demonstrating their utility for clustering areas into persistent, epidemic, and episodic transmission categories. However, the original analysis relied on custom scripts that were not packaged for reuse by the broader research community.

`epidesc` addresses this gap by encapsulating the epi-descriptors computation methodology into a modular and flexible R package. The target audience includes epidemiologists, data scientists, and public health analysts who need to systematically characterise disease transmission patterns from routine notification data. The package facilitates the reproduction and extension of epi-descriptors, allows comparative patterns across diseases and geographies, and supports the development of public health interventions tailored to local transmission patterns.

41 State of the field

42 Several R packages support epidemiological time series analysis. Early tools like surveillance
43 (Meyer et al., 2017) and EpiEstim (Cori et al., 2013) established standard frameworks for
44 outbreak detection and real-time reproduction number (R_t) estimation. Accelerated by the
45 COVID-19 pandemic, the wider EpiVerse-TRACE software initiative has significantly expanded
46 this toolkit. Modern packages include EpiNow2 (Abbott et al., 2026) for Bayesian estimation of
47 infection trajectories and R_t under reporting delays, incidence2 (Taylor, 2024) for computing
48 and visualising incidence curves, and specialized tools such as epiparameter (Lambert et
49 al., 2025) for epidemiological parameter extraction. Outside the R ecosystem, EpiViewer
50 (Thorve et al., 2018) provides a web-based framework for exploring, comparing, and organising
51 epidemiological time series, with built-in computation of general epidemic features such as
52 peak timing and magnitude, cumulative counts, and first take-off time. EpiPoi is an useful
53 epidemiological time series analysis tool for detecting trends, seasonality, anomalies and spatial
54 patterns (Alonso & McCormick, 2012). It was developed and compiled in MATLAB but its
55 last version is outdated. However, none of these packages provide a dedicated framework for
56 computing the specific set of temporal descriptors needed to characterise and classify disease
57 transmission patterns at a spatial unit.

58 The novelty of epidesc lies in its focus on descriptive features of epidemic time series rather
59 than on mechanistic modelling or outbreak detection. By computing indicators such as
60 the maximum duration of consecutive weeks with at least x cases, the proportion of weeks
61 exceeding this threshold, and the frequency of case-free periods, epidesc enables a data-driven
62 classification of transmission patterns. This approach complements existing tools and is
63 particularly relevant for diseases with strong seasonality and spatial heterogeneity, such as
64 dengue in tropical settings. Furthermore, epidesc is designed to accommodate the addition of
65 new descriptors from the research community.

66 Software design

67 epidesc is structured as a standard R package following CRAN conventions, with comprehensive
68 documentation via roxygen2 (Wickham et al., 2026), unit tests using testthat (Wickham,
69 2011), and a vignette demonstrating the complete workflow. The package depends on lubridate
70 (Grolemund & Wickham, 2011) for date manipulation and nseq (Saldanha, 2024) for efficient
71 run-length encoding operations on time series.

72 epidesc is available on CRAN [10.32614/CRAN.package.epidesc](https://cran.r-project.org/web/packages/epidesc/index.html) and the development version
73 can be accessed at <https://github.com/BSC-ES/epidesc>. The software architecture follows a
74 three-step pipeline (Figure 1):

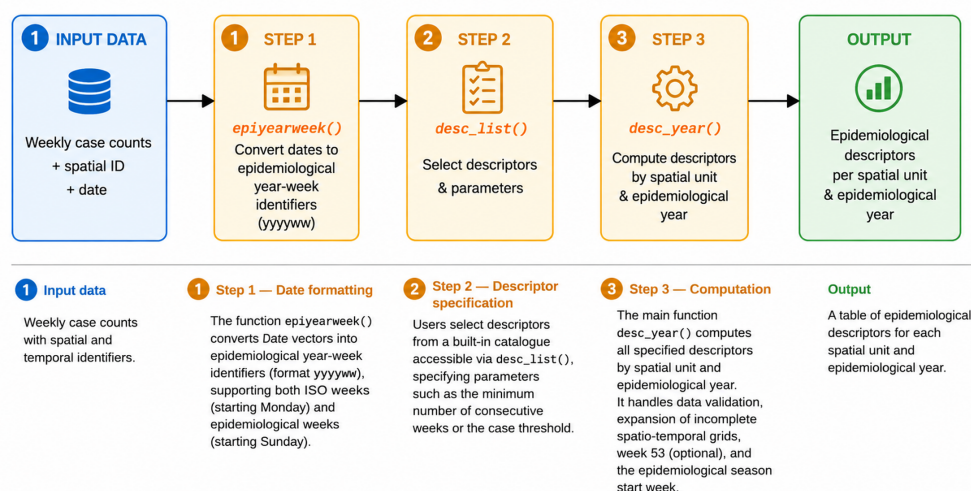


Figure 1: Schematic pipeline of the `epidesc` package. Input data containing weekly case counts with spatial and temporal identifiers are processed across three stages: date formatting with `epiyearweek()`, descriptor selection via `desc_list()`, and computation with `desc_year()`, producing epidemiological descriptors per spatial unit and epidemiological year.

Input data: As input, `epidesc` requires a dataframe containing epidemiological line-list data. This dataframe must include at least 3 columns: weekly case counts, a spatial identifier and a date.

Step 1 — Date formatting: the function `epiyearweek()` converts Date vectors into epidemiological year-week identifiers (format `yyyyww`), supporting both ISO weeks (starting Monday) and epidemiological weeks (starting Sunday).

Step 2 — Descriptor specification: users select descriptors from a built-in catalogue accessible via `desc_list()`, specifying parameters such as the minimum number of consecutive weeks or the case threshold. The catalogue currently includes 11 epi-descriptors organised into four classes (Table 1).

Step 3 — Computation: the main function `desc_year()` computes all specified epi-descriptors by spatial unit and epidemiological year.

Output: `epidesc` provides an object that stores selected epi-descriptors calculated by year or epidemiological year and spatial unit identifier.

Table 1- Epi-descriptors included in `epidesc` v0.1.0.

Class	Function	Description	Parameters
Peak	Ap	Maximum number of cases (peak) - amplitude	
Peak	Tp	Week where the maximum peak occurred - time	
Period with cases	Cnf	Frequency of periods of consecutive n weeks or longer with at least x cases	n, x
Period with cases	Cmax	Maximum duration in consecutive weeks with at least x cases	x

Class	Function	Description	Parameters
Period with cases	Cmed	Median duration in consecutive weeks with at least x cases	x
Period with cases	Isof	Number of weeks with isolated cases	
Period with cases	p	Proportion of weeks with at least x cases	x
Period without cases	Cwf	Frequency of periods of at least n consecutive weeks without cases	n
Period without cases	Cwmax	Maximum duration in consecutive weeks without cases	
Period without cases	Cwmed	Median duration in consecutive weeks without cases	
Incidence	Inc	Annual incidence per p population	p

90 With the exception of incidence rate calculations, the epi-descriptors do not require population
 91 estimates for the spatial unit of analysis. This facilitates their application in settings where
 92 population data are incomplete or not regularly updated, a common limitation at fine spatial
 93 scales and particularly relevant in resource-constrained settings.

94 The desc_year() function performs input validation by checking the regularity of the time series,
 95 identifying duplicate time points within spatial units, and verifying that population data are
 96 provided when incidence descriptor is requested. If the spatio-temporal grid is incomplete, the
 97 function issues a warning. Epi-descriptors are calculated only for complete epidemiological years;
 98 if an epidemiological year has missing weeks or case records, the corresponding descriptors are
 99 returned as NA. Week 53, when present, is optionally handled by equally distributing its cases
 100 between weeks 52 of the same year and 01 of the next year, ensuring consistent 52-week years
 101 for descriptor calculation. This behaviour is controlled by the argument collapse53, which is
 102 TRUE by default. A minimal usage example is as follows:

```
library(epidesc)
data("dengueRio")

# step 1: date formatting
dengue <- dengueRio
dengue$yearweek <- epiyearweek(dengue$date, start = "Sunday")

# step 2: descriptor specification
descriptors <- list(
  Ap = list(fun = "Ap"),
  Cnf3 = list(fun = "Cnf", n = 3, x = 5),
  Inc = list(fun = "Inc", p = 100000)
)

# step 3: computation of epi-descriptors
res <- desc_year(
  data = dengue,
  cases = "cases",
```

```
time = "yearweek",  
space = "muni_code",  
pop = "pop",  
sweek = 41,  
descriptors = descriptors  
)
```

The package includes a sample dataset (dengueRio) containing weekly dengue cases for municipalities in the state of Rio de Janeiro (2017–2022), enabling users to explore the functionality immediately after installation.

Research impact statement

The methodological framework implemented in `epidesc` was originally developed and applied in Almeida et al. (2022), which classified 1,823 Brazilian municipalities into four distinct dengue transmission patterns (persistent, epidemic, episodic/epidemic, and episodic) using data from 2010 to 2019. That study demonstrated the association between transmission patterns, population size, and climate, providing evidence for the existence of a critical community size for persistent dengue transmission. The classification framework has implications for the design of differentiated surveillance protocols and the achievement of World Health Organization goals for neglected tropical disease elimination.

By packaging this methodology into `epidesc`, we enable its application to other diseases (e.g., Zika, chikungunya, malaria), other geographies, and longer time periods. The package is currently being used within the Global Health Resilience (GHR) group at the Barcelona Supercomputing Center in Spain and at the Oswaldo Cruz Foundation (Fiocruz) in Brazil for ongoing research on climate-sensitive disease dynamics and early warning systems. The open-source availability of `epidesc` facilitates reproducibility and encourages community contributions of additional descriptors.

Future developments and call for contributions

In future versions of the package, we intend to expand the range of epi-descriptors to include methodologies to characterise the seasonality of the disease. Furthermore, we call for contributions to extend the list of supported epi-descriptors and welcome any suggestions either via Github issues <https://github.com/BSC-ES/epidesc/issues> or by sending an email to the corresponding authors.

AI usage disclosure

AI tools were used to code-review the package (in addition to human code review), to assist with unit testing and support in English review.

Author contributions

RML, IFA, and CTC conceived and implemented epi-descriptors and curated the data demo. CMG developed the R package. RML, CMG, and DSL contributed to package design. RML, CMG, DSL, and DRX conducted tests and package validation. RML, CMG, IFA, DSL, DRX and RL reviewed package GitHub repository. RML and DRX wrote the manuscript. All authors reviewed the final manuscript version.

Conflict of interest

We declare we have no competing interests.

Acknowledgements

Raquel M. Lana is funded by the European Union (Marie Skłodowska-Curie Actions, grant agreement 101109642).

Diego Ricardo Xavier and Raquel M. Lana acknowledge the project CNPq 445194/2024-3 - Development of Indicators for the Detection and Characterization of Anomalies in Climate-Sensitive Tropical Diseases in Brazil.

Rachel Lowe, Claudia T. Codeço and Christovam Barcellos acknowledge the Wellcome Trust HARMONIZE 224694/Z/21/Z.

Rachel Lowe and Claudia T. Codeço acknowledge the Wellcome Trust IDExtremes 226069/Z/22/Z.

Iasmim Ferreira de Almeida acknowledges the Wellcome Trust (Mosqlimate 226088/Z/22/Z) and the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES, Finance Code 001).

Christovam Barcellos and Diego Ricardo Xavier acknowledge the financial support from CNPq (process 444665/2023-4) and the Pasteur Network (Climate-Health Observatory Accelerator Project).

Daniela Lührsén was supported by the Barcelona Supercomputing Center AI4Science Fellowship programme funded by the Recovery and Resilience Mechanism-Next Generation as part of the Spanish Ministry's Recovery, Transformation and Resilience Plan.

References

- Abbott, S., Hellewell, J., Sherratt, K., Gostic, K., Hickson, J., Badr, H. S., DeWitt, M., Azam, J. M., Lison, A., & Funk, S. (2026). *EpiNow2: Estimate and forecast real-time infection dynamics* (Version v1.9.0). Zenodo. <https://doi.org/10.5281/zenodo.20728851>
- Alcayna, T., Rao, V. B., & Lowe, R. (2025). Identifying the climate sensitivity of infectious diseases: A conceptual framework. *The Lancet Planetary Health*, 9(8), 101291. <https://doi.org/10.1016/j.lanplh.2025.101291>
- Almeida, I. F. de, Lana, R. M., & Codeço, C. T. (2022). How heterogeneous is the dengue transmission profile in Brazil? A study in six Brazilian states. *PLOS Neglected Tropical Diseases*, 16(9), e0010746. <https://doi.org/10.1371/journal.pntd.0010746>
- Alonso, W. J., & McCormick, B. J. (2012). EPIPOI: A user-friendly analytical tool for the extraction and visualization of temporal parameters from epidemiological time series. *BMC Public Health*, 12(1). <https://doi.org/10.1186/1471-2458-12-982>
- Cori, A., Ferguson, N. M., Fraser, C., & Cauchemez, S. (2013). A new framework and software to estimate time-varying reproduction numbers during epidemics. *American Journal of Epidemiology*, 178(9), 1505–1512. <https://doi.org/10.1093/aje/kwt133>
- Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. *Journal of Statistical Software*, 40(3), 1–25. <https://doi.org/10.18637/jss.v040.i03>
- Lambert, J. W., Kucharski, A., & Tamayo, C. (2025). *Epiparameter: Classes and helper functions for working with epidemiological parameters* (Version v0.4.1). Zenodo. <https://doi.org/10.5281/zenodo.14748221>

- 179 Meyer, S., Held, L., & Höhle, M. (2017). Spatio-temporal analysis of epidemic phenomena
180 using the r package surveillance. *Journal of Statistical Software*, 77(11), 1–55. <https://doi.org/10.18637/jss.v077.i11>
181
- 182 Saldanha, R. (2024). *Nseq: Count of sequential events*. <https://doi.org/10.32614/CRAN.package.nseq>
183
- 184 Tabataba, F. S., Chakraborty, P., Ramakrishnan, N., Venkatramanan, S., Chen, J., Lewis, B.,
185 & Marathe, M. (2017). A framework for evaluating epidemic forecasts. *BMC Infectious*
186 *Diseases*, 17(1). <https://doi.org/10.1186/s12879-017-2365-1>
- 187 Taylor, T. (2024). *incidence2: Compute, handle and plot incidence of dated events*. <https://CRAN.R-project.org/package=incidence2>
188
- 189 Thorve, S., Wilson, M. L., Lewis, B. L., Swarup, S., Vullikanti, A. K. S., & Marathe, M. V.
190 (2018). EpiViewer: An epidemiological application for exploring time series data. *BMC*
191 *Bioinformatics*, 19(1). <https://doi.org/10.1186/s12859-018-2439-0>
- 192 Wickham, H. (2011). Testthat: Get started with testing. *The R Journal*, 3, 5–10. <https://journal.r-project.org/articles/RJ-2011-002/>
193
- 194 Wickham, H., Danenberg, P., Csárdi, G., & Eugster, M. (2026). *roxygen2: In-line*
195 *documentation for r*. <https://doi.org/10.32614/CRAN.package.roxygen2>
- 196 Xavier, D. R., Magalhães, M. de A. F. M., Gracie, R., Reis, I. C. dos, Matos, V. P. de, &
197 Barcellos, C. (2017). Difusão espaço-tempo do dengue no município do rio de janeiro,
198 brasil, no período de 2000-2013. *Cadernos de Saúde Pública*, 33(2). <https://doi.org/10.1590/0102-311x00186615>
199